Count the Overlap Once

Unit 2 · Topic 2.5 · TODAY'S PROBLEM

Suppose 120 students answer two yes-or-no questions: "Do you like pizza?" and "Do you like tacos?" The table summarizes their answers. Let P be the event that a randomly selected student likes pizza and T the event that the student likes tacos.

Jamie: "$P(P \text{ or } T) = 80/120 + 60/120 = 140/120$. Because a probability cannot exceed 1, this must mean everyone likes at least one; I would report the answer as 1."

Noor: "Some students may have been counted in both totals, so we need the intersection before deciding."

Student food preferences (counts)

Preference	Pizza	Not pizza	Total
Tacos	35	25	60
Not tacos	45	15	60
Total	80	40	120

FIRST TAKE (5 min, on your sheet)

Does the table support reporting everyone likes pizza or tacos? Choose a claim and cite one count.

DOK 1 **(a)** State the number in $P \cap T$ and write its probability as a fraction.

DOK 2 **(b)** Find the probability that a selected student likes at least one food, counting each student once. Also find the probability of liking neither.

DOK 3 **(c)** Are the two events mutually exclusive, and whose reasoning is supported? Use the overlap and your probability to explain Jamie's error. Explain what reporting everyone would hide. Write 3–4 sentences.

Video + follow-along: `u4_lesson3-4-5_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.